

Bounded Relational Capacity and Its Conditional Continuum Dynamics

Born–Infeld Candidates, Signature Selection, and Open Projection Bridges

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Abstract. We organize the bounded-relaxation approach to emergent dynamics around a single structural axiom. The capacity axiom [A-cap] states that the per-node relaxation rate of an admissible relational configuration is uniformly bounded by a constant c_{BI} . The axiom is motivated by non-injective projection together with finite local distinguishability; it is not a consequence of non-injectivity alone, which constrains information rather than rates. Which continuum quantity inherits the bound is an open problem: null field configurations carry arbitrarily large amplitudes at vanishing invariants, so no local Lagrangian of the field invariants bounds every constitutive direction. Born–Infeld theory is accordingly presented as a saturation candidate, distinguished by its known exceptional-propagation properties, not as a derived or unique completion. Under explicit hypotheses — an established continuum limit for the relational Laplacian and the existence of a second-order hyperbolic sector — the admissible signature is Lorentzian $(-+++)$, and stationary spherically symmetric flux conservation yields the Newtonian exterior profile $\Phi = \Phi_0 - C/r$. The Schwarzschild form requires additional dynamical input, and the reading of horizons as capacity saturation is relative to a preferred relaxation foliation; both are stated as conditional or open rather than as theorems. **Interpretive status.** The paper provides a structural scaffold connecting relational capacity to effective continuum dynamics. Its physical content becomes predictive only when the identified projection bridges — from capacity to constitutive response, from relaxation to hyperbolic dynamics, and from symmetric connectivity to gauge curvature — are closed.

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1 Introduction

A growing body of work explores the possibility that space-time geometry is not a fundamental structure but an effective description emerging from more primitive relational or operator-based systems. In several approaches, discrete relational data, weighted graphs, or spectral operators admit continuum limits in which geometric notions such as distance, dimension, and curvature arise as derived quantities rather than postulated ones [5]. Within the same programme, non-injective effective descriptions have been analyzed as a source of intrinsic limits on observable factorization [1], and a pre-geometric relational framework has been proposed as the underlying substrate [3].

The present paper isolates the load-bearing structural input of the bounded-relaxation approach and states precisely what follows from it, under which hypotheses, and what remains open. The input is a single axiom: admissible relational configurations relax at a uniformly bounded per-node rate. We call this the capacity axiom, [A-cap], and we denote the bound c_{BI} , a notation that records

the Born–Infeld candidate which motivates it and which companion papers of the programme use as an input.

The paper makes five statements, each with an explicit status. First, [A-cap] is formulated as an axiom on a named relational variable, motivated — but not implied — by non-injective projection (Section 2). Second, the bridge between the discrete capacity bound and any bounded continuum constitutive quantity is an open problem, and we explain why no local Lagrangian of the field invariants can close it by itself (Section 3). Third, Born–Infeld theory is a saturation candidate distinguished by its exceptional-propagation properties, and the derivation of such a propagation condition from projection structure is the open route to any future selection statement (Section 4). Fourth, under an established continuum limit for the relational Laplacian and under the hypothesis of a second-order hyperbolic sector, the effective signature is Lorentzian; the hypothesis is necessary, because bounded-flux transport admits finite-speed parabolic realizations (Sections 5–7). Fifth, stationary spherically symmetric flux conservation yields the Newtonian exterior profile; the Schwarzschild

form and the saturation reading of horizons require additional input and are presented as conditional statements and interpretation, not as results (Sections 8 and 9).

2 The capacity axiom

2.1 Formulation

Let $G = (V, E, w)$ be a locally finite weighted graph with symmetric weights $w_{ij} = w_{ji} \geq 0$, let $\phi : V \rightarrow \mathbb{R}$ be a relational configuration, and let τ be the monotonic ordering parameter of the relaxation process introduced in Section 5. The named relational variable of this paper is the *per-node relaxation rate* $|\partial_\tau \phi_v|$.

Axiom [A-cap] (bounded relational capacity). There exists a constant $c_{\text{BI}} > 0$ such that, for every admissible configuration and every node $v \in V$,

$$|\partial_\tau \phi_v| \leq c_{\text{BI}}. \quad (1)$$

The axiom is stated at the discrete relational level, where the variable it bounds is unambiguous. Any continuum reformulation requires the bridges discussed in Section 3 and is not part of the axiom.

2.2 Structural motivation from non-injective projection

Let $\Pi : \Omega \rightarrow \mathcal{O}$ denote the projection from the relational configuration space Ω to effective observables $\mathcal{O}$, and let $\mathcal{E}_y = \Pi^{-1}(y)$ denote the fibre over $y \in \mathcal{O}$. When Π is non-injective, the fibres are non-trivial and the effective description cannot resolve arbitrarily fine relational distinctions [1].

Non-injectivity alone, however, does not bound any rate. A projection may discard information while the retained variables evolve arbitrarily fast: for $\Pi(x, y) = x$ on $\mathbb{R}^2$, the fibres are entire lines and nothing constrains $\partial_\tau x$. The motivation for (1) therefore requires a second ingredient: *finite local distinguishability*, the assumption that an effective cell supports only finitely many projectively distinguishable states per relaxation step. Under finite local distinguishability and locality of the relational dynamics, an unbounded per-node rate would transport more distinctions per step than the effective description can register, and the excess would be unresolvable. This chain motivates the axiom; it does not derive it, because finite local distinguishability is itself an assumption about the projection. [A-cap] is accordingly an axiom, not a theorem.

A related structural argument concerns gauge structure. If Π is injective and compatible with relational locality, each observable configuration lifts uniquely, no non-trivial re-identification of fibres exists, and no emergent internal symmetry would be available on this account; emergent gauge structure would therefore require non-injectivity. We use this argument only as motivation for studying non-injective projections. Its rigorous formulation requires constructing the projected bundle and its connection, which is

part of the framework programme [3] and is not attempted here.

2.3 What the axiom does not assert

[A-cap] does not assert that any continuum field strength, constitutive response, energy flux, or curvature is bounded. It does not select an effective Lagrangian. It does not fix the relation between c_{BI} and any continuum scale, including the saturation scale β of Section 4. Each of these connections is a separate bridge with its own status, examined below.

3 The open constitutive bridge

Suppose a continuum regime exists in which slowly varying effective fields admit a local action

$$S = \int d^4x \mathcal{L}(S, P), \quad S = \frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad P = \frac{1}{4} F_{\mu\nu} \tilde{F}^{\mu\nu}, \quad (2)$$

with Maxwell weak-field limit $\mathcal{L} \simeq -S$. The question raised by [A-cap] is which continuum quantity, if any, inherits the discrete capacity bound.

The candidate identifications differ and are inequivalent: a bound on the field strength F , a bound on the constitutive response $D^{\mu\nu} = \partial \mathcal{L} / \partial F_{\mu\nu}$, a bound on the invariants (S, P) , and a bound on an energy flux are four distinct conditions. No local Lagrangian of the invariants bounds every direction at once: null configurations satisfy $S = P = 0$ at arbitrarily large field amplitude, so any $\mathcal{L}(S, P)$ with a Maxwell limit leaves the response in null directions unbounded. Born–Infeld theory itself bounds the purely electric field strength through the reality domain of its square root, while leaving null amplitudes unconstrained; it does not bound F , D , and all invariants indiscriminately.

Consequently, no selection statement of the form “bounded capacity forces Lagrangian X ” can even be formulated until the bounded continuum quantity is specified and derived from the projection structure. We record this as the first open bridge.

Open Problem 1 (constitutive bridge). Derive from the projection structure which effective continuum quantity inherits the capacity bound (1), and in which domain of field configurations the inherited bound is active.

The relaxation-side counterpart of this bridge is the passage from the first-order relaxation flow of Section 5 to a second-order effective dynamics supporting propagating modes; it is stated as a hypothesis in Section 7.

4 The Born–Infeld candidate

Among nonlinear completions of Maxwell theory, the Born–Infeld Lagrangian [8]

$$\mathcal{L}_{\text{BI}} = \beta^2 \left(1 - \sqrt{1 + \frac{2S}{\beta^2} - \frac{P^2}{\beta^4}} \right) \quad (3)$$

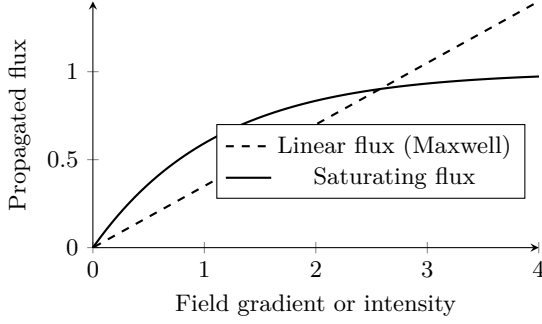


Fig. 1. Linear versus saturating flux response. A saturating response motivates completions of Born–Infeld type but does not by itself select one; the selection criterion established in the literature is exceptional propagation, not saturation.

is the historical prototype of a saturating theory: it reduces to Maxwell theory for $|S|, |P| \ll \beta^2$, its reality domain bounds the purely electric field strength by β , and it assigns a finite self-energy to the point charge [8, 11]. In covariant determinantal form, with an effective metric $g_{\mu\nu}$,

$$S_{\text{eff}} = \beta^2 \int d^4x \left(\sqrt{-\det(g_{\mu\nu})} - \sqrt{-\det\left(g_{\mu\nu} + \frac{1}{\beta} F_{\mu\nu}\right)} \right). \quad (4)$$

Within the present framework, (4) is a *candidate*: it is compatible with several partial notions of saturation, and it is the candidate whose known structural properties best match the intent of [A-cap] (Figure 1). It is not derived here, and it is not unique with respect to any bounded-quantity condition formulated so far. Distinct completions realize distinct partial notions of saturation — families with a bounded reality domain other than the square root, and Lagrangians with saturating response at unbounded invariants, both exist — so that, pending Open Problem 1, no selection theorem is available to state.

What genuinely distinguishes Born–Infeld theory in the nonlinear electrodynamics literature is its *exceptional propagation*: among nonlinear local theories $\mathcal{L}(S, P)$ with a Maxwell limit, absence of birefringence — a single characteristic cone for both photon polarizations — singles out the Born–Infeld Lagrangian, Maxwell theory itself being the trivial birefringence-free case [7, 13]. This is a theorem under hypotheses about wave propagation, not about bounds. If the projection framework is to select (4), the structural route indicated by the literature is therefore to derive an exceptional-propagation condition from the projection of relational characteristics.

Open Problem 2 (propagation bridge). Determine whether the projection structure implies a single-characteristic-cone condition on the effective dynamics; if it does, the Born–Infeld candidate is selected by the Boillat–Plebański theorem, and the capacity bound acquires a derived constitutive meaning.

5 Relational graph, Laplacian operator, and relaxation dynamics

We consider an underlying relational system represented by a weighted graph $G = (V, E, w)$, where V denotes a set of nodes, E a set of edges, and $w_{ij} \geq 0$ encodes the strength of the relation between nodes i and j . No embedding in a background spacetime is assumed. The graph is taken as a purely relational object, encoding adjacency and coupling structure between abstract degrees of freedom.

A scalar field ϕ_i is defined on the nodes of the graph. The fundamental operator governing relational variations is the discrete Laplacian,

$$(L\phi)_i = \sum_j w_{ij} (\phi_i - \phi_j), \quad (5)$$

which measures the local mismatch of ϕ with respect to its relational neighborhood. Throughout this work the graph is locally finite and the weights are symmetric, $w_{ij} = w_{ji}$.

The system is assumed to admit an irreversible relaxation dynamics driven by the Laplacian,

$$\frac{d\phi_i}{d\tau} = -(L\phi)_i, \quad (6)$$

where τ is a monotonically increasing parameter labeling the progression of the relaxation process, subject to the capacity axiom (1). No interpretation of τ as a fundamental physical time is imposed; it serves as an ordering parameter associated with the irreversible flow toward admissible stationary configurations, and the existence of such an ordering distinguishes one direction of evolution from its reverse.

Equation (6) is first order in τ . It defines a relaxation flow, not a wave dynamics: by itself it supports no oscillatory propagating modes and no finite characteristic cone. Whether and how a second-order effective dynamics arises from the relaxation flow is not settled in this paper; the signature analysis of Section 7 therefore proceeds from an explicit hypothesis rather than from (6).

Stationary configurations satisfy $L\phi = 0$ and correspond to relational equilibria. Slowly varying configurations describe regimes in which the system admits an effective coarse-grained description whose large-scale observables depend primarily on the spectral properties of the Laplacian rather than on the detailed microscopic structure of the graph. The graph structure does not imply fundamental discreteness of physical space or time; different microscopic realizations with the same large-scale spectral properties are physically equivalent within the effective description.

6 Conditional continuum limit

We consider a sequence of weighted graphs $G_N = (V_N, E_N, w^{(N)})$ with $|V_N| \rightarrow \infty$, satisfying (i) uniformly bounded degree growth relative to the intrinsic scaling, (ii) local quasi-isotropy of the weighted second moments of adjacency in the low-energy sector, and (iii) sufficient decay of $w_{ij}^{(N)}$

with intrinsic relational distance. The working hypothesis of this section is that, within a fixed low-energy spectral window, the discrete Laplacians converge in the strong resolvent sense to a second-order differential operator

$$\mathcal{L} = \nabla_\mu (A^{\mu\nu}(x) \nabla_\nu) \quad (7)$$

on a smooth manifold, with $A^{\mu\nu}$ a symmetric tensor arising from the second weighted moments of adjacency. The framework-specific formulation of this limit is developed in [5]; its status there is that of a reconstruction under stated regularity and density hypotheses, and the present paper treats it as a conditional input.

The established mathematical results closest to this hypothesis concern graphs sampled from, or quasi-uniformly discretizing, a Riemannian manifold: pointwise and spectral convergence of graph Laplacians is proved in that setting [6, 9]. These theorems presuppose the manifold whose geometry is recovered. An intrinsic convergence statement — one that does not presuppose a background manifold, as the relational programme requires — is not available in the literature and is open within the programme. We record the dependence explicitly: every geometric statement below is conditional on the continuum limit.

Under the hypothesis (7), the principal symbol

$$\sigma_2(\mathcal{L})(x, k) = A^{\mu\nu}(x) k_\mu k_\nu \quad (8)$$

provides a coordinate-independent definition of an effective inverse metric, $g^{\mu\nu}(x) \propto A^{\mu\nu}(x)$, up to a choice of units [5]. The effective metric is a derived object: distinct microscopic graphs with the same principal symbol are indistinguishable at the level of effective geometry, and the continuum description is defined only up to spectral equivalence within the selected energy window.

One structural limitation must be stated explicitly. The present symmetric weighted Laplacian does not construct an antisymmetric gauge curvature; a magnetic or bundle-valued graph Laplacian would be additional structure. The gauge field strength $F_{\mu\nu}$ appearing in Section 4 is therefore not derived from (5); its relational origin is a separate construction problem.

Open Problem 3 (gauge-sector bridge). Construct a relational operator — necessarily carrying more structure than the symmetric weighted Laplacian — whose continuum limit produces an antisymmetric curvature two-form coupled to the effective metric, and determine whether the capacity axiom constrains it.

7 Conditional signature selection

This section derives the effective signature from explicit hypotheses.

Hypothesis [H-hyp] (second-order sector). In the homogeneous low-energy regime, the effective dynamics admits a sector governed by a second-order-in- τ equation of motion whose principal symbol is a non-degenerate quadratic form $A_0^{\mu\nu} k_\mu k_\nu$ on frequency-momentum space

$(k_0, \mathbf{k})$, and effective propagation is required to exhibit a finite characteristic cone with τ as the single ordering direction.

[H-hyp] is a hypothesis, for two reasons stated in Sections 5 and 3. First, the underlying relaxation flow (6) is first order in τ , and the passage to a second-order sector is not constructed here. Second, the capacity axiom does not by itself force hyperbolicity: bounded-flux transport admits finite-speed *parabolic* realizations. Flux-limited (tempered) diffusion equations propagate fronts at finite speed while remaining degenerate parabolic [14]; they are, moreover, structurally of Born-Infeld type. A finite characteristic cone therefore does not follow from the capacity bound alone and must be demanded of the sector under consideration.

Under [H-hyp], the signature is fixed, up to overall sign of $A_0^{\mu\nu}$, by a standard trichotomy. A definite $A_0^{\mu\nu}$ (signature $(++++)$) yields an elliptic operator: no finite characteristic cone exists. A signature with more than one time-like direction yields an ultra-hyperbolic operator, for which the Cauchy problem is not well posed in the classical Hadamard sense [12]. The only remaining possibility is the Lorentzian case, for which the Cauchy problem is well posed [10]: exactly one negative direction, aligned with the ordering parameter τ . After normalization of the maximal propagation speed to $c = 1$, the homogeneous effective metric is

$$g_{\mu\nu} = \eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1). \quad (9)$$

The scope of this statement is exactly its hypotheses. Conditional on the continuum limit (Section 6) and on [H-hyp], homogeneity selects flat space with Lorentzian signature, and the null cone $ds^2 = 0$ marks the maximal admissible effective propagation rate, in structural parallel with causal saturation in Born-Infeld-type theories [7, 11]. Unconditionally, nothing in this section asserts that the relational substrate produces a hyperbolic sector; that is the relaxation-side bridge left open in Section 3.

8 Stationary radial profiles under spherical flux conservation

We now consider regimes in which homogeneity is broken by a localized, stationary, spherically symmetric obstruction: a region of enhanced relational stiffness whose microscopic origin is left unspecified. Outside the obstruction the effective operator of Section 6 takes the form

$$\mathcal{L} = \nabla_\mu (A^{\mu\nu}(r) \nabla_\nu), \quad A^{\mu\nu}(r) = \text{diag}(-A_t(r), A_r(r), A_\perp(r), A_\perp(r)) \quad (10)$$

with the sign structure inherited from Section 7.

For a stationary scalar probe Φ of the effective structure, the exterior equation $\nabla_\mu (A^{\mu\nu} \nabla_\nu \Phi) = 0$ reduces to radial flux conservation,

$$r^2 A_r(r) \frac{d\Phi}{dr} = C, \quad (11)$$

with C an integration constant measuring the strength of the obstruction. If the exterior region is asymptotically homogeneous, so that $A_r(r) \rightarrow 1$ with the normalization of Section 7, the leading exterior profile is

$$\Phi(r) = \Phi_0 - \frac{C}{r} + O\left(\sup_{r' \geq r} |A_r(r') - 1| C/r\right). \quad (12)$$

This is the Newtonian exterior profile. It is the robust content of this section: it follows from stationarity, spherical symmetry, flux conservation, and asymptotic normalization, all conditional on the continuum limit.

The Schwarzschild geometry does not follow from (12). A general static spherically symmetric effective line element,

$$ds^2 = -f_t(r) dt^2 + f_r(r) dr^2 + r^2 d\Omega^2, \quad (13)$$

is constrained by the weak-field identification $f_t = 1 + 2\Phi$ only in its temporal coefficient. The Schwarzschild form requires in addition the closure $f_t f_r = 1$, which is not a coordinate choice: it is a dynamical condition, equivalent in the effective-metric language, given the asymptotic normalization, to the vacuum statement $R^t_t = R^r_r$, and metrics violating it — for instance $f_t = 1 - r_s/r$ with $f_r = 1$ — satisfy every constraint used in this section. Within the programme, dynamical input of this type is supplied, conditionally on independently matched coefficients, by the spectral stationarity route of the companion gravity papers [2, 4]. We therefore state the conditional result only: under the hypotheses of this section and, as external dynamical input, the closure $f_t f_r = 1$ together with the exact temporal identification $f_t = 1 + 2\Phi$ with $\Phi_0 = 0$, the exterior effective geometry is Schwarzschild with $r_s = 2C$ in relational units; with the weak-field identification alone, the Schwarzschild form holds only to leading order in C/r . No claim is made that the present framework derives the closure or the exact identification.

9 Discussion, status ledger, and outlook

9.1 Status ledger

The statements of this paper have the following statuses.

- **Axiom.** [A-cap], the uniform bound (1) on the per-node relaxation rate. Companion papers of the spectral admissibility programme use this bound as an input; stating it as a named axiom places their derivations on an explicit, named hypothesis.
- **Structural motivation (not derivation).** Non-injectivity, spectral convergence of graph Laplacians is established for graphs built on an underlying manifold [6, 9]; the relational programme requires an intrinsic counterpart, and until one exists the geometric statements here remain conditional on the framework-specific limit of [5]. In transport theory, flux-limited diffusion [14] realizes bounded flux with finite-speed fronts in a parabolic setting, demonstrating that a capacity bound does not entail hyperbolicity and thereby fixing the honest status of [H-hyp] as a hypothesis.
- **Computational.** The Maxwell weak-field limit of the Born–Infeld candidate, its electric-domain saturation, and the finite point-charge self-energy are classical results [8, 11].
- **Conditional.** Lorentzian signature selection, given the continuum limit and [H-hyp] (Section 7). The Newtonian exterior profile (12), given the continuum limit, stationarity, spherical symmetry, and asymptotic normalization. The Schwarzschild form, given additionally the closure $f_t f_r = 1$ from external dynamical input.
- **Open.** The constitutive bridge (Open Problem 1), the propagation bridge and any Born–Infeld selection statement (Open Problem 2), the gauge-sector bridge (Open Problem 3), the first-order-to-second-order passage underlying [H-hyp], and the intrinsic (embedding-free) continuum limit.

9.2 Horizons as foliation-relative saturation: an interpretation

In the conditional Schwarzschild exterior, the static chart adapted to the relaxation ordering degenerates at $r = r_s$: the temporal coefficient f_t vanishes while curvature invariants remain finite. This degeneracy is chart-relative — charts regular across r_s exist, and in them the effective Cauchy problem continues — so no invariant breakdown of the effective operator occurs at the horizon. If, however, the relaxation ordering τ defines a physically preferred foliation, as the relational framework suggests, then the degeneracy of the τ -adapted chart acquires structural meaning: the horizon is the locus where the preferred relational slicing ceases to be transverse to effective propagation. Reading this locus as *capacity saturation* would additionally require a derived relation between the local flux scale C/r_s^2 and the capacity bound c_{BI} , which is not available. A complementary reading in terms of projection entropy — the growth of fibre multiplicity limiting local resolvability — is developed in the non-injectivity analysis [1]. Both readings are interpretation, not results, and we record them as outlook.

9.3 Relation to neighbouring programmes

Three bodies of work delimit the present contribution. In nonlinear electrodynamics, the selection theorems of Boillat and Plebański [7, 13] single out the Born–Infeld Lagrangian among nonlinear completions by exceptional propagation; the present programme shares the goal of structural selection but must *derive* the selecting condition from projection structure rather than postulate it, which is precisely Open Problem 2. In discrete-to-continuum analysis, spectral convergence of graph Laplacians is established for graphs built on an underlying manifold [6, 9]; the relational programme requires an intrinsic counterpart, and until one exists the geometric statements here remain conditional on the framework-specific limit of [5]. In transport theory, flux-limited diffusion [14] realizes bounded flux with finite-speed fronts in a parabolic setting, demonstrating that a capacity bound does not entail hyperbolicity and thereby fixing the honest status of [H-hyp] as a hypothesis.

9.4 Interpretive outlook

Interpretive status. The structural reading of this paper is that a single finite capacity, if it governs the relational substrate, would leave three recognizable signatures in effective physics: a saturating constitutive response, a preferred Lorentzian causal structure, and saturation-limited strong-field regions. The paper shows that each signature currently stands at the end of an open bridge — constitutive, dynamical, and geometric respectively — and formulates the bridges precisely. Closing any one of them would convert the corresponding signature from a motivated reading into a derived consequence; failing to close them would confine the capacity axiom to the discrete relational level. This is the calibrated sense in which bounded relational capacity may, but does not yet, explain why saturation structures pervade effective physics.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used generative AI tools as an auxiliary support for language refinement, structural clarification, and consistency checking. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the scientific content, interpretations, and conclusions of the published article.

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